

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Digital Devices Mock Examination

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Total Marks: 200 Marks | Time Focus: Standard Syllabus Pacing

Question 1 (Total: 25 marks)

A local library is upgrading its computer networks and hardware devices to improve services for visitors.

(a) State the definition of a **digital device**. (1)

Definition

(b) Identify one input peripheral device from the list below that the library could use to record physical book pages as digital images. Place a cross [X] in the correct box. (1)

☐ **A.** OMR Scanner

☐ **B.** Barcode Scanner

☐ **C.** Flatbed Document Scanner

☐ **D.** Graphics Tablet

(c) The library uses an automated security system. Describe how an **input peripheral sensor** can function automatically within this system without human intervention. (2)

Description of automated process

(d) Visitors at the library bring their own portable devices. Explain one benefit of a device having **lower energy consumption** characteristics. (2)

Explanation of operational benefit

(e) The library installs 'all-in-one' desktop computers for research stations. Describe the physical hardware configuration of an **all-in-one desktop computer**. (2)

Hardware layout parameters

(f) Some researchers require large-scale architectural maps to be printed directly from Computer Aided Design (CAD) software.

(i) Identify the specialized vector-drawing output peripheral needed for this task. (1)

Peripheral device name

(ii) Describe how this device physically operates to draw images on expansive sheets of paper. (2)

Mechanical operation sequence

(g) The library administrator uses a specialized system to execute advanced background scripts. The system utilizes a **Command Line Interface (CLI)**. Explain two benefits of using a CLI instead of a Graphical User Interface (GUI). (4)

Benefit 1 & Benefit 2 structural blocks

(h) The library updates its registration system to include high-security biometric scanners.

(i) State **two physical human characteristics** that can be read by biometric scanners. (2)

1. / 2.

(ii) Explain why **iris recognition** is considered significantly safer and more accurate than standard fingerprint scanning. (2)

Comparative structural security argument

(i) The main library office operates a large server room requiring complex statistical calculations and bulk processing of city-wide data sets. Identify the type of large, powerful multi-user computer required for this application and state its primary use. (6)

Type of computer framework / Statement of primary deployment use cases

Question 2 (Total: 30 marks)

Modern office workers rely heavily on mobile telecommunication and media display devices.

(a) Every mobile smartphone requires a SIM card to establish network connections.

(i) State the full meaning of the acronym **SIM**. (1)

Full nomenclature string

(ii) State the primary function of a SIM card when it is inserted into a mobile phone. (1)

Primary operational role

(b) Smartphones are excellent examples of **multifunctional devices**.

(i) Define the term *multifunctional device*. (2)

Technical standard definition

(ii) List three distinct hardware functions combined within a smartphone that historically required separate electronic systems. (3)

1. / 2. / 3.

(c) Because smartphones incorporate multiple advanced hardware systems, their power profiles are affected. Explain two design features that cause smartphones to have shorter battery lives than older, basic mobile phones. (4)

Feature 1 & Feature 2 technical design factors

(d) A company designs specialist mobile units, such as the Alto 2 'talking phone'. Identify the specific target group for the Alto 2 phone and explain how its design factors ensure accessibility. (3)

Target Demographic Group / Explanation of acoustic accessibility controls

(e) Tablet computers are often used by employees on the move. State two features that tablets share with smartphones and one physical hardware difference between them. (3)

Shared features / Architectural hardware differences

(f) When purchasing new computer screens, an office administrator reviews the resolution properties.

(i) Define the term **resolution** regarding display technology. (1)

Technical metric definition

(ii) Explain the visual degradation effect known as a **pixelated display**. (2)

Visual degradation mechanism parameters

(g) Employees frequently stream training videos online. Contrast the digital management process of **streaming** media content with traditional **downloading**. (4)

Volumetric buffer streaming paths vs. Local file preservation memory layers

(h) Identify one wireless network standard used to transmit media from a phone to a smart TV over short distances. Place a cross (X) in the correct box. (1)

☐ A. Linear Barcode

☐ B. Bluetooth

☐ C. Microcontroller

☐ D. ROM

Question 3 (Total: 30 marks)

Computer architecture relies on solid processor performance and precise memory management to function effectively.

(a) Define the term **performance** as it applies to internal computer systems. (1)

System clock and throughput metric boundaries

(b) Central Processing Units (CPUs) work alongside Random Access Memory (RAM). Describe the individual roles of the processor and RAM during an active software execution cycle. (4)

Processor core execution paths / RAM system execution parameters

(c) When a user runs too many programs simultaneously, system RAM may run out of available bytes.

(i) State the name of the storage allocation area used by the system when RAM space is fully occupied. (1)

Memory expansion workspace name

(ii) Explain how this extension memory functions, stating its physical location within secondary storage structures. (3)

Staging functionality / Secondary drive tracking coordinates

(d) A laptop specification sheet lists properties for Expansion Capability. Describe how an engineer can use slots or ports to update a system's **expansion capability**. Provide an example of a common connection standard. (3)

Description of hardware extension via bus interfaces / Example common standard (e.g., USB-C, PCIe)

(e) Distinguish between a **file** and a **program**. (2)

Comparative structural differentiation

- (f) Security features help keep private user data safe from unauthorized access. Contrast the functionality of a device **lock command** with a device **wipe command**. (4)

Access restriction boundaries vs. Permanent cryptographic storage sanitization loops

- (g) Complete the following table by inserting the correct definitions for the specified User Interface types. (4)

User Interface Type	Functional Definition (Sourced from Textbook Guidelines)
Menu-Driven Interface	
Gesture Interface	

- (h) Voice Interfaces frequently capture audio commands from users using speech recognition tools. Explain why a Voice Interface device often needs a continuous internet connection during runtime operations. (2)

Acoustic modeling analytics / Cloud parser dependencies

- (i) Identify two common household appliances that are controlled by internal microprocessors. (2)

1. / 2.

Question 4 (Total: 35 marks)

A school ICT lab requires a comprehensive update to its data capture and printing peripherals.

- (a) Define what is meant by an **input peripheral**. (1)

Syllabus base definition

- (b) Laptop computers feature integrated touch-sensitive control pads to save workspace. State the technical term for this specific pointing device feature. (1)

Technical identifier nomenclature

- (c) A student uses a stylus with a flat interactive drawing pad for digital art creation. Identify this input device combination from the syllabus list. (1)

System combination name

- (d) Optical technology is extensively deployed across modern logistics and tracking infrastructure.

- (i) Describe how a barcode scanner reads data lines. (2)

Illumination reflective laser diagnostics

- (ii) Differentiate between standard linear barcodes and two-dimensional matrix codes (QR codes) in terms of capacity and scanning flexibility. (3)

Data storage bounds vs. Multidirectional grid capture flexibility

- (e) Optical scanners run software packages to convert physical traits into editable data records. Compare the primary tasks of **OCR** and **OMR** software engines. (4)

Character raster identification layers vs. Mark presence boolean checks

- (f) Touch screens are manufactured using two primary technological concepts: Resistive and Capacitive. Compare resistive and capacitive touch screen technology. Your response must reference physical activation methods, screen durability, and multi-touch calculations. (6)

Pressure mapping vs. Electrostatic charge variations / Surface resiliency parameters

- (g) A school music room is installing microphones to register acoustic sound paths. Describe how a microphone transforms physical sound waves into digital signals via its internal components. (3)

Diaphragm induction mechanics / ADC translation resolution loops

- (h) Automated manufacturing setups use physical parts known as actuators to modify environments.
(i) Define the term **actuator** (control device). (2)

Functional control definition boundaries

- (ii) Give two real-world physical examples of actuators. (2)

1. / 2.

- (i) The design lab installs three-dimensional physical printing arrays. Explain how a 3D printer constructs real-life objects from solid material filament feeds. (3)

Additive cross-sectional staging mechanics

(j) A computer monitor's operational configuration is evaluated by specific technical markers. State three factors used to evaluate the selection of display monitor components. (3)

1. / 2. / 3.

Question 5 (Total: 40 marks)

Data retention requires reliable non-volatile staging solutions, scaling formulas, and secure media parameters.

(a) Define the term **secondary storage** and explain why it must be non-volatile. (2)

Syllabus tracking definition

(b) Explain the conceptual difference between a **storage device** and a **storage medium**. (2)

Drive mechanics vs. Physical layer medium diagnostics

(c) Hard Disk Drives (HDDs) use mechanical parts to access data paths.

(i) Describe the physical design of an HDD. Your answer must mention platters and read/write heads. (2)

Magnetic structure physical configuration properties

(ii) Define the term **seek time** as it relates to magnetic disk drives. (1)

Mechanical latency boundaries

(d) Optical storage media formats carry differing maximum limits due to tracking densities. State the exact maximum storage capacity value for each format below: (3)

Compact Disc (CD) / Digital Versatile Disc (DVD) / Blu-ray Disc limits

(e) Flash memory devices (Solid-State) are highly recommended for mobile hardware operations. Explain why Flash media is uniquely robust and suitable for portable items compared to mechanical hard disks. (3)

Solid state alignment benefits / Absence of kinetic drop targets

(f) Identify the linear secondary storage media form used primarily for large-scale whole-system backing up and archiving in enterprise environments. (1)

Syllabus structural medium entry

(g) International GCSE ICT scaling criteria use a standard base-1000 multiplier configuration system.

(i) State the total number of bits contained in exactly one byte. (1)

Bit counts

(ii) State the exact byte count equivalent to 1 Megabyte (MB). (1)

Base-1000 standard layout size value

(h) Show your working for the following data capacity conversion tasks:

(i) A media project file occupies 3.8 Gigabytes (GB) of disk space. Convert this storage volume into Megabytes (MB) using base-1000 multipliers. (2)

Calculation workflow workspace blocks

(ii) A secure system vault archives data sizing up to 4,500,000 Kilobytes (KB). Convert this volume capacity metric into Gigabytes (GB). (3)

Multi-stage resolution layout tracks

(i) Complete the following structural matrix regarding storage characteristics using data from the textbook. (12)

Media Format Type	Relative Data Access Speed	Maximum Capacity Scale	Cost per Gigabyte Metric	Portability Profile
Flash Media	Fastest	Varies, up to 128 GB	Very High	
Hard Disk		Varies, up to many TB		Not suitable
Optical Media	Slow		Medium	
Magnetic Tape		Varies, up to 185 TB	Low	Not suitable

(j) A legacy computing manual states that Blu-ray media drives show backwards compatibility. Explain backwards compatibility with reference to Blu-ray and DVD optical discs. (3)

Laser wavelength adjustments / Diagnostic structural playback paths

Question 6: Extended Response Essay (Total: 25 marks)

A video game development studio wants to create an interactive virtual reality application. The software tracks physical motion in real-time, displays graphical avatars on screen, and connects users across global networks. The technical directors must choose the appropriate user hardware layout. They are evaluating two options:

- **Option 1:** Traditional high-performance desktop computer bases combined with upgradeable internal components and wired peripherals.
- **Option 2:** Compact, portable handheld console rigs that feature converged, multifunctional architectures.

Evaluate both options. You should discuss the impacts of each option on processing performance, storage capacity, expansion capability, and ease of deployment for the studio's development teams. (25)

Structured Technical Essay Response Block (Include structural evaluation arguments and a formal concluding synthesis metric)

End of Question Bank